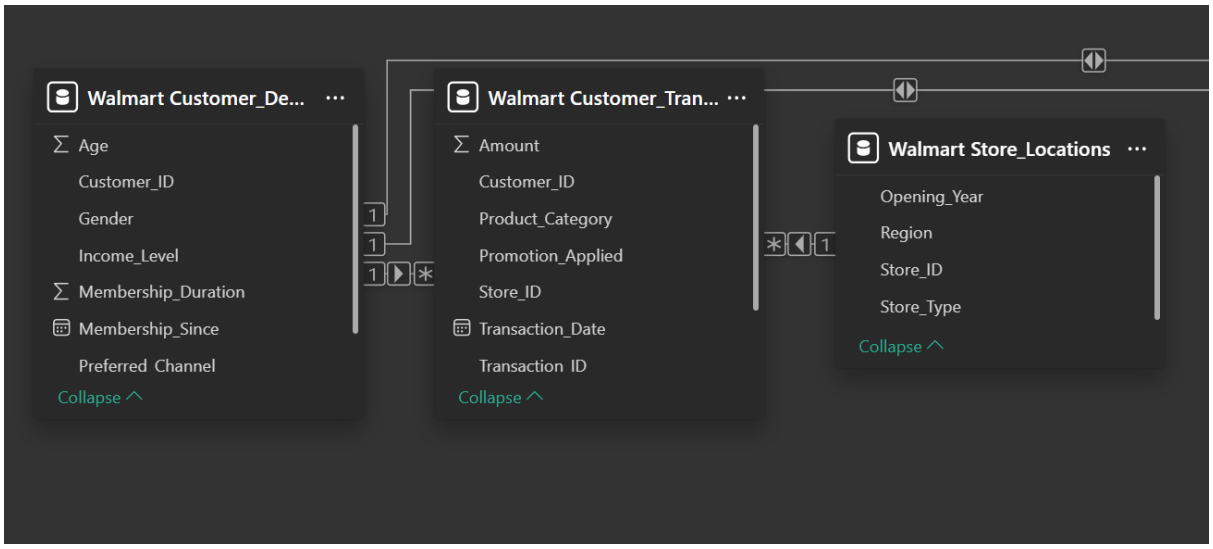
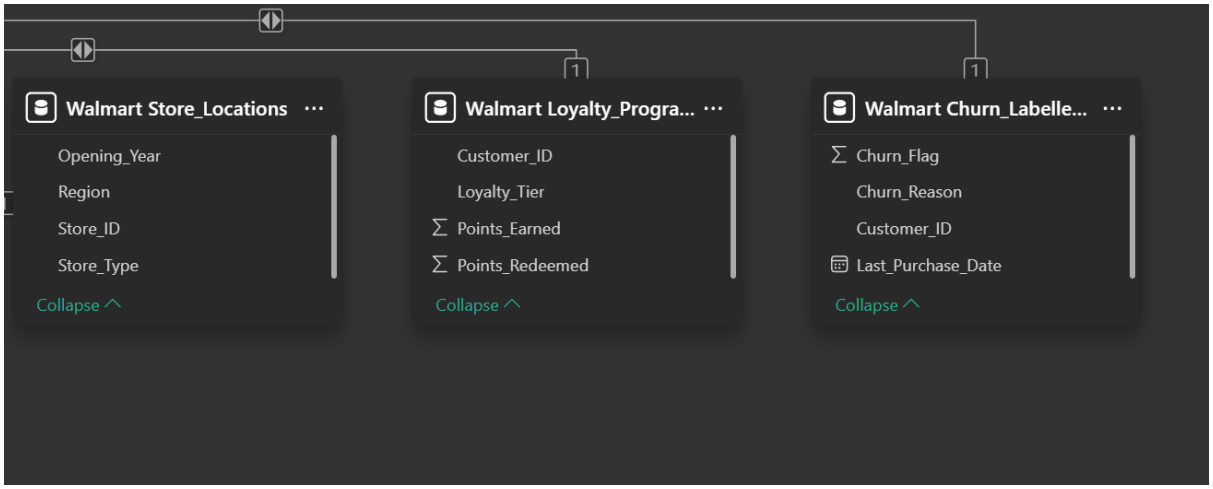
Video Link:

https://drive.google.com/file/d/1OhLj1NJ_ygfpdAdu8olVZSWN06PiHs9E/view?usp=sharing

Task 1: Data Modelling & Cleaning

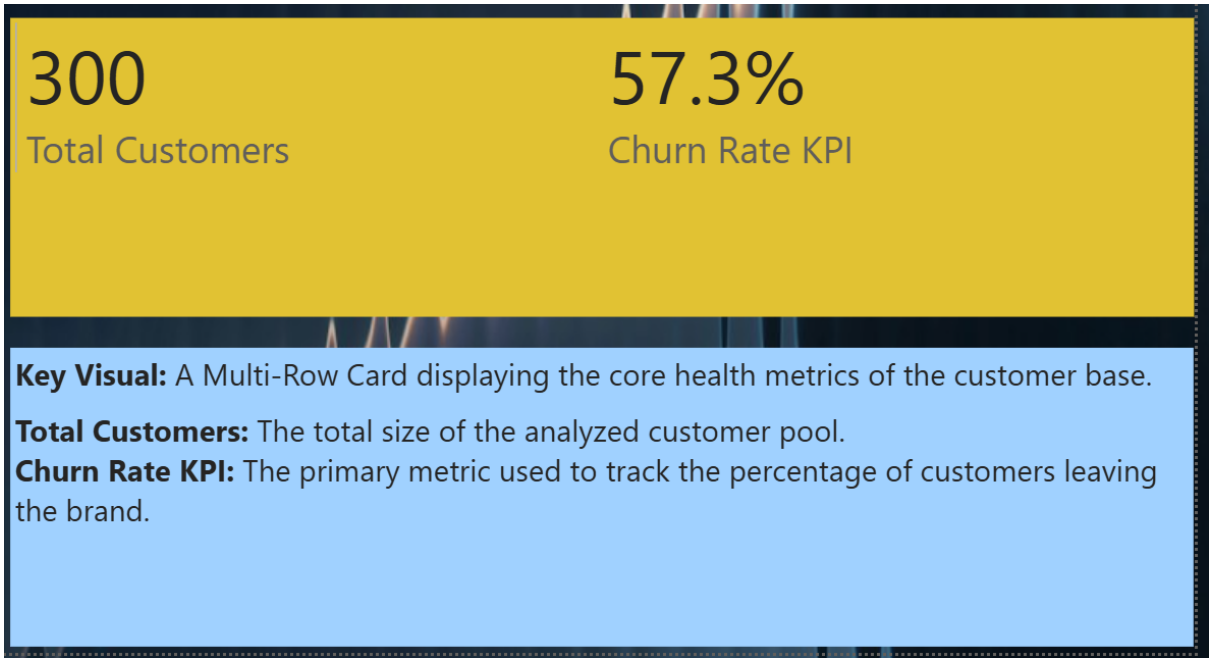
Customer_ID	Age	Gender	Region	Income_Level	Membership_Since	Preferred_Channel	Membership_Duration
CUST0001	19	Female	Central	High	30 December 2022	Online	1091
CUST0003	21	Female	South	Medium	10 September 2025	Online	106
CUST0008	63	Female	Central	Medium	05 September 2024	Online	476
CUST0009	42	Female	Central	Low	04 March 2019	Online	2488
CUST0012	37	Female	South	High	27 November 2023	Store	759
CUST0017	41	Female	Central	Medium	30 October 2020	Store	1882
CUST0022	31	Female	West	Low	31 January 2024	Store	694
CUST0024	31	Female	South	Medium	03 September 2023	Store	844
CUST0031	68	Female	North	High	03 June 2024	Online	570
CUST0032	40	Female	South	Low	14 October 2021	Store	1533
CUST0033	55	Female	East	Medium	10 July 2022	Online	1264
CUST0035	21	Female	North	Medium	21 May 2025	Online	218
CUST0039	24	Female	Central	High	26 January 2021	Online	1794
CUST0040	47	Female	South	Medium	17 June 2020	Store	2017
CUST0042	45	Female	West	High	17 May 2023	Online	953
CUST0043	54	Female	East	Low	25 August 2021	Online	1583
CUST0045	20	Female	East	Medium	27 October 2020	Store	1885
CUST0050	64	Female	East	Low	15 May 2018	Store	2781
CUST0055	41	Female	West	Low	04 September 2017	Store	3034
CUST0059	56	Female	South	High	11 November 2022	Online	1140





Transaction_ID	Customer_ID	Store_ID	Product_Category	Transaction_Date	Amount	Pro
TXN00005	CUST0209	STORE041	Groceries	29 March 2024	941.99	Yes
TXN00015	CUST0005	STORE037	Groceries	19 April 2024	852.34	No
TXN00020	CUST0145	STORE030	Groceries	24 May 2024	689.04	Yes
TXN00037	CUST0272	STORE045	Groceries	05 February 2024	137.5	Yes
TXN00059	CUST0116	STORE046	Groceries	09 May 2024	591.17	Yes
TXN00066	CUST0132	STORE005	Groceries	14 July 2024	759.61	Yes
TXN00073	CUST0026	STORE021	Groceries	15 March 2024	408.12	No
TXN00074	CUST0295	STORE007	Groceries	19 June 2024	289.74	Yes
TXN00095	CUST0196	STORE026	Groceries	10 September 2024	692	No
TXN00099	CUST0041	STORE045	Groceries	18 March 2024	662.25	Yes
TXN00118	CUST0272	STORE005	Groceries	25 November 2024	375.09	Yes
TXN00119	CUST0027	STORE007	Groceries	10 April 2024	775.26	Yes
TXN00126	CUST0099	STORE025	Groceries	16 December 2024	743.12	Yes
TXN00128	CUST0015	STORE017	Groceries	09 April 2024	631.15	No
TXN00150	CUST0128	STORE037	Groceries	27 May 2024	337.59	No
TXN00151	CUST0042	STORE045	Groceries	25 February 2024	539.38	No

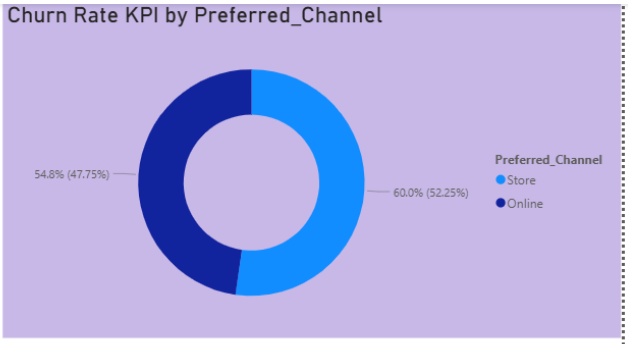
Task 2: Churn & Retention Metrics

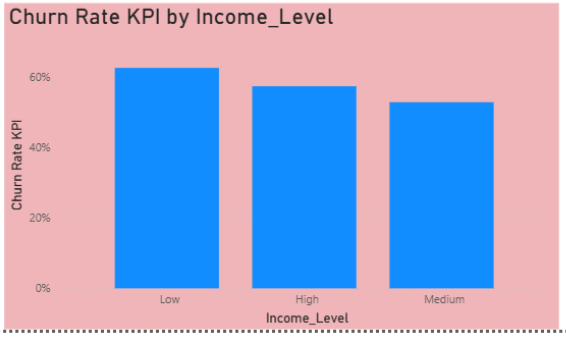


Key Visual: A Multi-Row Card displaying the core health metrics of the customer base.

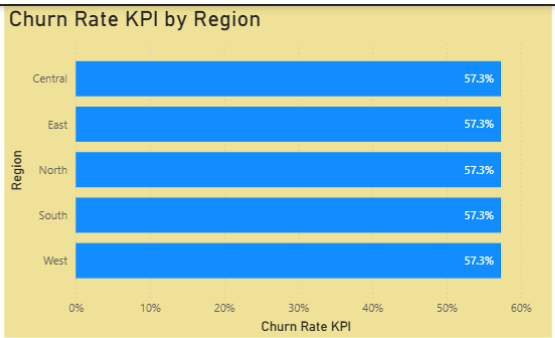
Total Customers: The total size of the analyzed customer pool.
Churn Rate KPI: The primary metric used to track the percentage of customers leaving the brand.

Donut Chart: Visualizes the relationship between Preferred Channels (e.g., Online, In-store) and churn, showing which shopping methods have the highest customer loyalty.



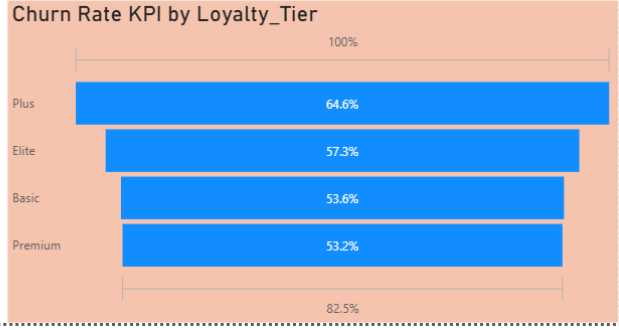


Clustered Column Chart: Analyzes churn based on Income Level, helping to determine if specific economic segments are more prone to leaving.



Clustered Bar Chart: Compares the Churn Rate KPI across different Regions, allowing users to identify specific geographical areas where retention is a problem.

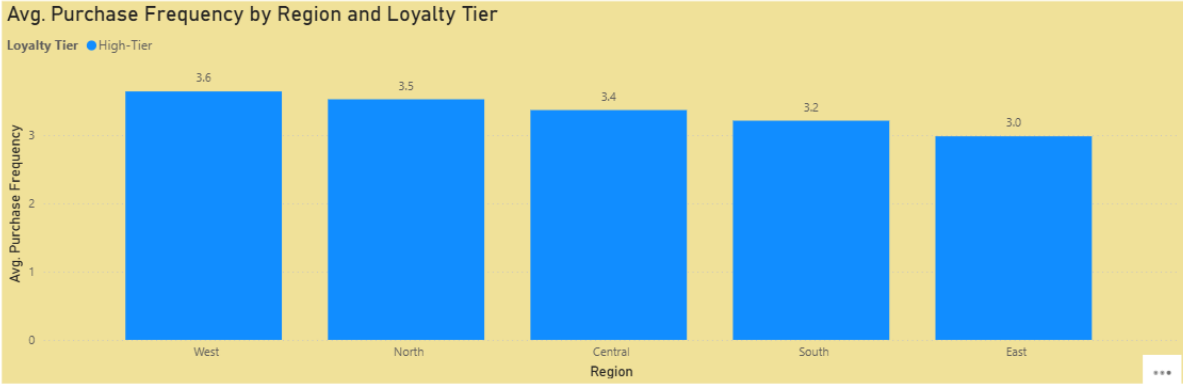
Funnel Chart: Uses the Loyalty Tier (e.g., Plus, Elite, Basic) to show how churn rates improve as customers move deeper into the loyalty program.



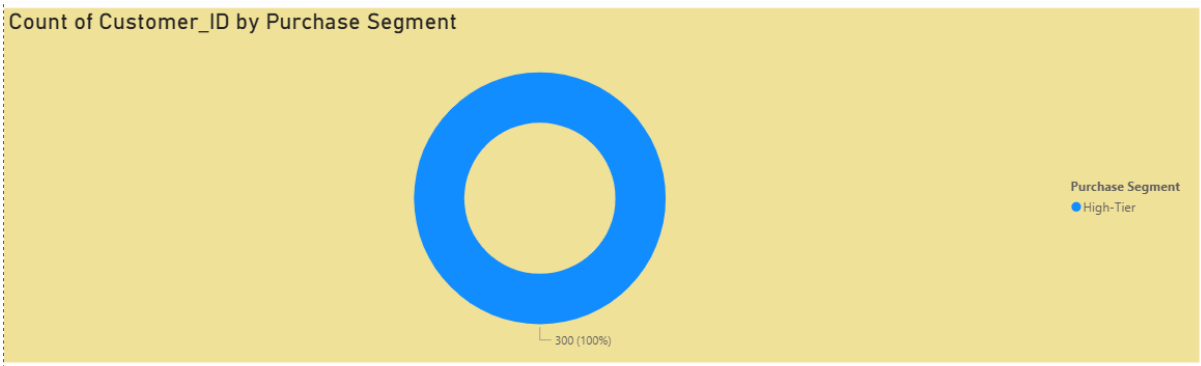

```
Churn Rate KPI = DIVIDE(COUNTROWS(FILTER('Walmart Churn_Labelled_Customers',[Days since Last Purchase]
>421)),COUNTROWS('Walmart Churn_Labelled_Customers'))
```

```
Days since Last Purchase = DATEDIFF('Walmart Churn_Labelled_Customers'[Last_Purchase_Date], TODAY(),
DAY)
```

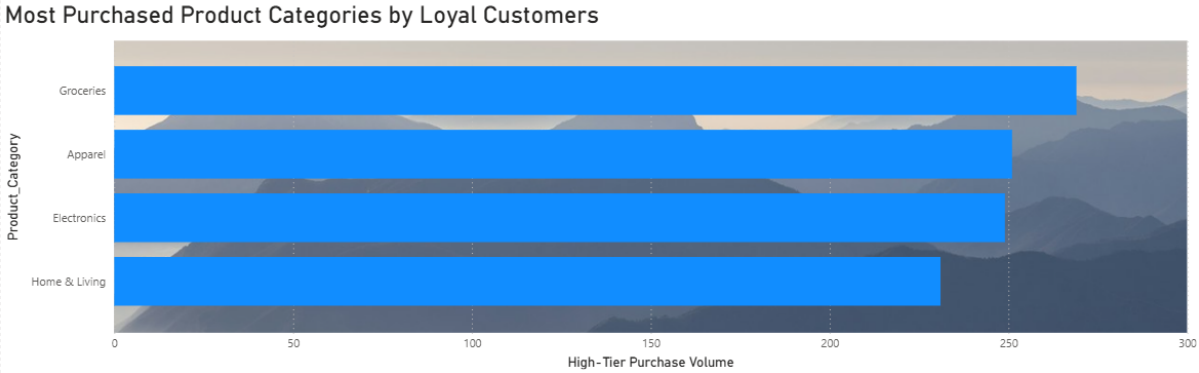
Task 3: Repeat Purchase Analysis



Key Visual: Clustered Column Chart
Region: Breaks down data by geographic location.
Loyalty Tier: Segments the data by membership status (e.g., High, Low).
Avg. Purchase Frequency: Measures how often customers shop.
Purpose: To determine if higher loyalty tiers actually correlate with higher shopping frequency and how this pattern varies by region.



Key Visual: Donut Chart
Purchase Segment: Categorizes customers based on their buying habits.
Customer Count: Shows the distribution of the total customer base across these segments.
Purpose: To understand the share of different customer types (e.g., occasional vs. frequent shoppers).



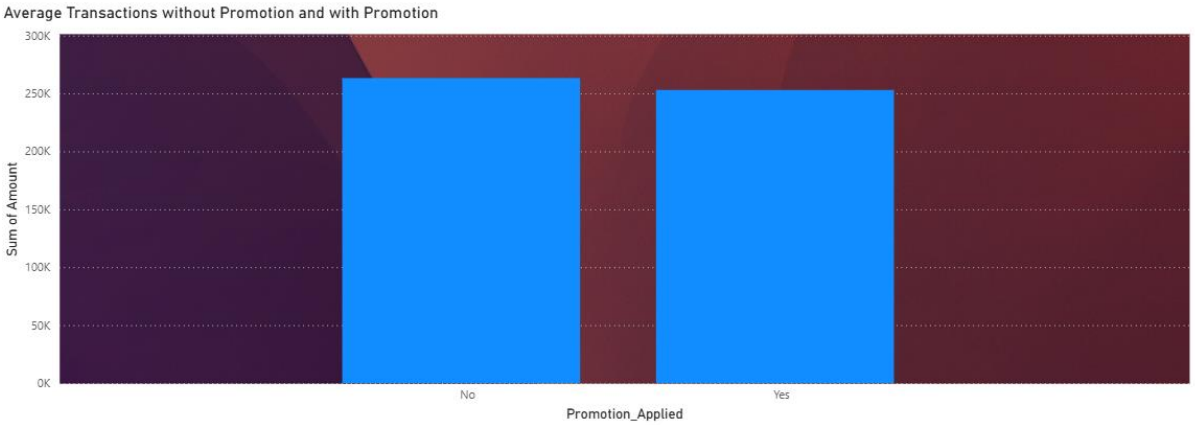
Key Visual: Clustered Bar Chart
Product Category: Identifies specific departments or goods (e.g., Electronics, Grocery).
High-Tier Purchase Volume: Focuses specifically on the spending volume generated by the most loyal or high-spending segments.
Purpose: To pinpoint which product categories are most popular with Walmart's most valuable customers.

Task 4: Promotion & Loyalty Impact

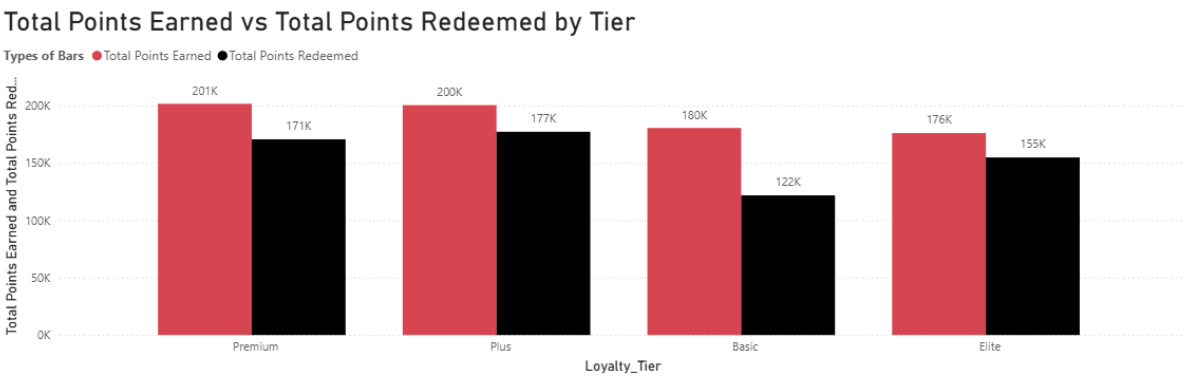
49.00%

Percentage of Transactions with Promotion

Key Visual: Card
Metrics: % Transactions with Promotion
Purpose: To quickly see the adoption rate of promotional offers across all transactions.



Key Visual: Clustered Column Chart
Promotion Applied: Compares transactions with promotions vs. those without.
Total Revenue: Shows the total sales volume for each group.
Purpose: To determine if promotions are successfully driving higher total revenue or if they are primarily being used on lower-value purchases.



Key Visual: Clustered Column Chart
Loyalty Tier: Segments the data by membership level (e.g., Plus, Elite, Basic).
Total Points Earned vs. Total Points Redeemed: A dual-measure comparison for each tier.
Purpose: To identify engagement gaps. For example, if a tier earns many points but redeems few, it may indicate that the rewards offered are not attractive enough for that specific group.

Recommendations to improve redemption & retention:-

Tier-Specific Catalysts: For "Premium" members with high point balances, move away from generic discounts. Implement exclusive access rewards (e.g., early access to new product lines or "VIP-only" shopping hours).

Localized Rewards: If the "Midwest" region shows high churn but high interest in the "Grocery" category, offer points-redemption options for essential services like "Free Grocery Delivery" specifically for that region.

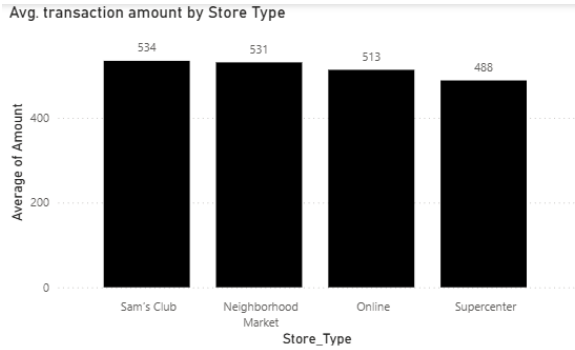
Promote High-Frequency Categories: Use promotions primarily on categories that drive repeat visits (like Groceries or Personal Care) rather than one-time purchases.

```
Percentage of Transactions with Promotion = AVERAGEX('Walmart Customer_Transactions',
IF('Walmart Customer_Transactions'[Promotion_Applied] = "Yes", 1, 0))
```

```
Total Points Earned = SUM('Walmart Loyalty_Program'[Points_Earned])
```

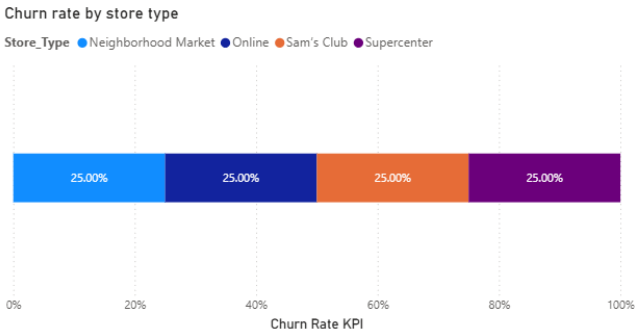
```
Total Points Redeemed = SUM('Walmart Loyalty_Program'[Points_Redeemed])
```

Task 5: Store & Channel Performance vs Retention



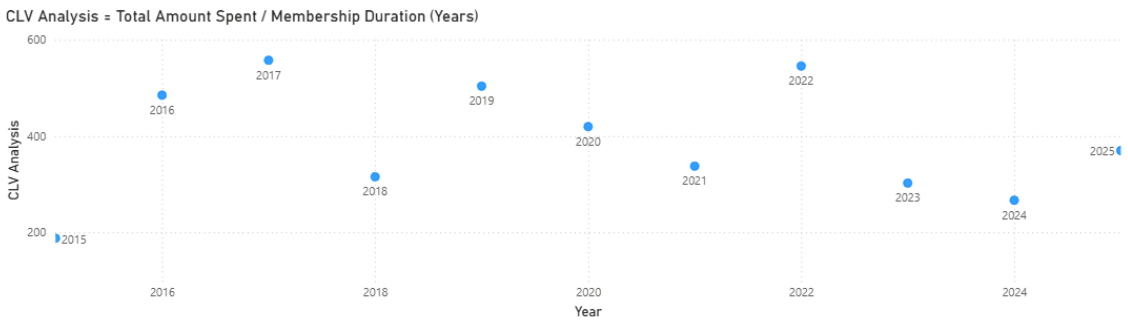
Visual Type: Clustered Column Chart
Metrics: Sum of Amount (Revenue) vs. **Store Type** (e.g., Supercenter, Neighborhood Market, Express).
Purpose: To identify which store format is the primary revenue driver for the company and how revenue is distributed across different physical footprints.

Visual Type: 100% Stacked Bar Chart
Metrics: Churn Rate vs. **Store Type**.
Purpose: This chart allows for a relative comparison. For example, it helps determine if "Price" is a bigger churn reason at "Neighborhood Markets" compared to "Supercenters," or if "Service Quality" is a recurring issue in one specific store format.



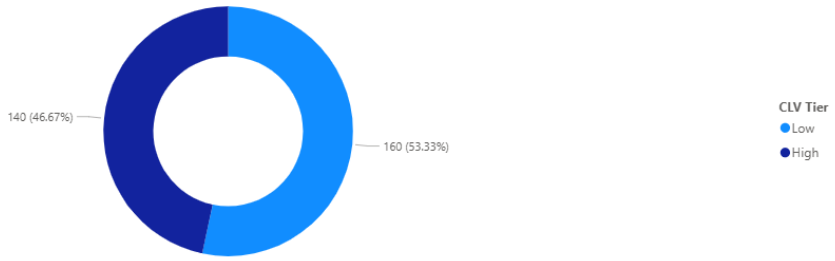
Promotion_Applied	Transaction_Month	Transaction_Year	Walmart Store_Locations.Region	Merged Store and Transactions.Opening_Year
Yes	03	2024	South	2017
No	04	2024	South	2001
Yes	06	2024	South	1992
No	08	2024	South	1993
Yes	07	2024	South	2000
Yes	03	2024	South	2000
Yes	02	2024	South	2004
Yes	10	2024	South	1992
Yes	05	2024	South	2000
Yes	08	2024	South	2018
Yes	02	2024	South	2011
No	08	2024	South	2011
No	09	2024	South	1992
Yes	06	2024	South	2000
No	05	2024	South	2003
Yes	01	2024	South	2004
No	01	2024	South	2001
No	04	2024	South	1994
Yes	04	2024	South	1994

Task 6: Customer Lifetime Value (CLV) Analysis

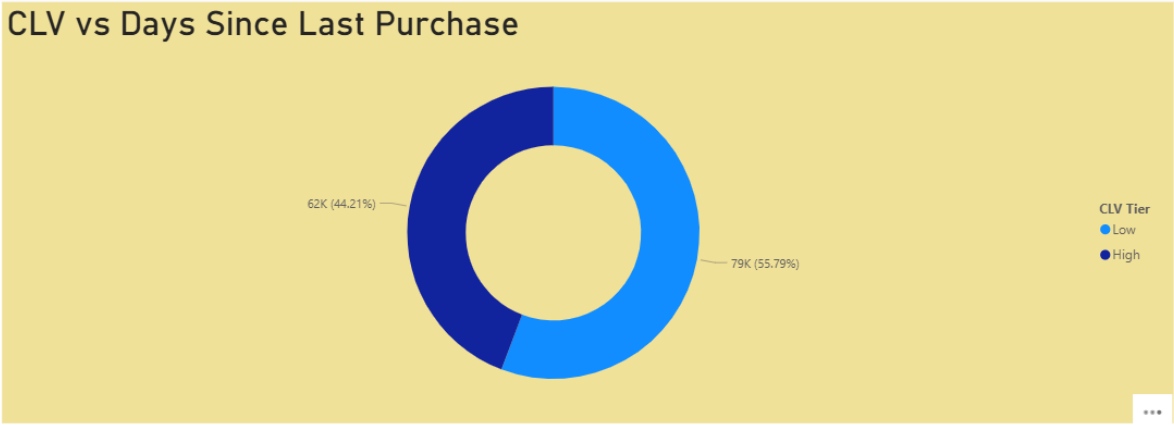


Key Visual: Scatter Chart
Metrics: CLV Analysis plotted against the Membership_Since date hierarchy (Year, Quarter, Month, Day).
Purpose: To identify if specific customers (e.g., those who joined in a specific year) are significantly more valuable than others over time.

Segmented customers into Low, High CLV

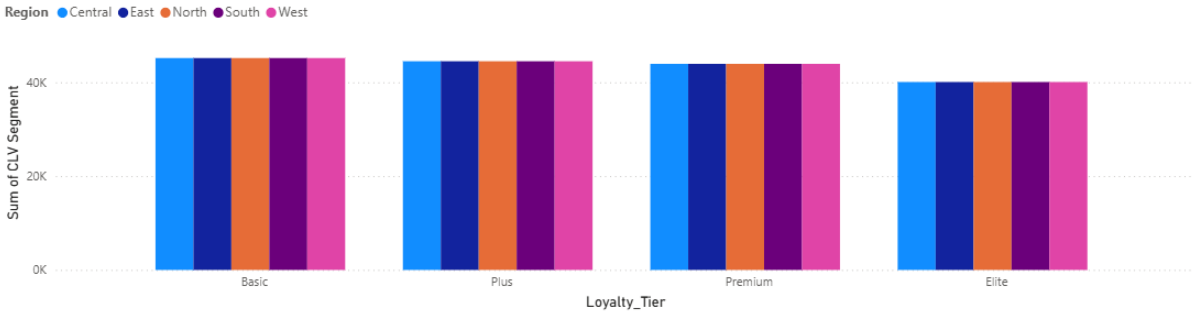


Key Visual: Donut Chart
Metrics: Customer Count vs. CLV Tier.
Purpose: To visualize the "Value Pyramid." It shows what percentage of the customer base falls into high-value vs. low-value tiers, helping prioritize marketing resources toward the most profitable segments.



Key Visual: Donut Chart
Metrics: Days Since Last Purchase vs. CLV Tier.
Purpose: This is a "Recency" analysis. If high-value CLV tiers have a large sum of "Days Since Last Purchase," it indicates that Walmart's most valuable customers are becoming inactive and are at high risk of churning.

CLV vs Loyalty_Tier and Region



Key Visual: Clustered Column Chart
Metrics: CLV Segment analyzed by Region and Loyalty Tier.
Purpose: To determine which regions are most successful at cultivating high-value loyalty members. It highlights where the loyalty program is effectively translating into high lifetime value.

```
CLV Segment = VAR TOTALAMOUNT = SUM('Walmart Customer_Transactions'[Amount])
VAR MEMBERDURATION = 'Walmart Customer_Demographics'[Membership_Duration]
RETURN
DIVIDE(TOTALAMOUNT,MEMBERDURATION,0)
```

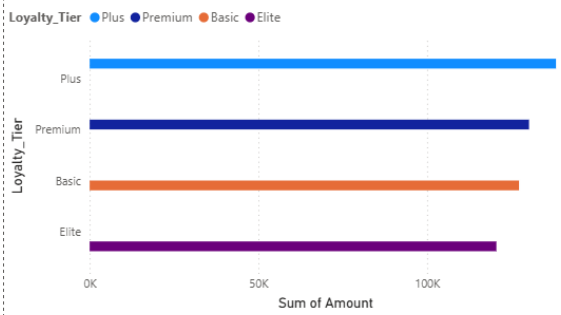
```
CLV Tier =
VAR IndividualCLV = [CLV Segment]
VAR AverageCLV = AVERAGEX(ALL('Walmart Customer_Demographics'[Customer_ID]), [CLV Segment])
RETURN
IF(IndividualCLV >= AverageCLV, "High", "Low")
```

Task 7: Final Dashboard & Executive Summary

57.3%	85.4%	$4,130.1
Churn Rate	Repeat Purchase Rate %	CLV Analysis

Key Visual : Uses a Multi-Row Card to display the three essential measures.
Metrics: Repeat Purchase Rate %, CLV Analysis, and Churn Rate.

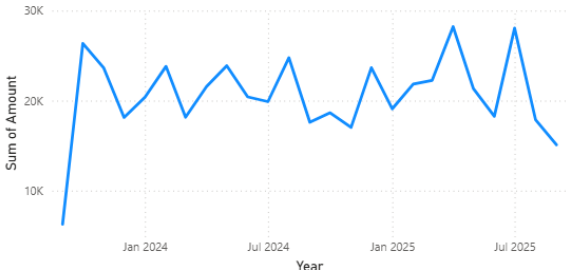
Loyalty & Promotion Impact



Trends & Tiers: a Clustered Bar Chart showing revenue contributions by Loyalty Tier.

Trends & Tiers : Features a Line Chart to track sales over time (Month/Year)

Loyalty & Promotion Impact Amount by Year and Month



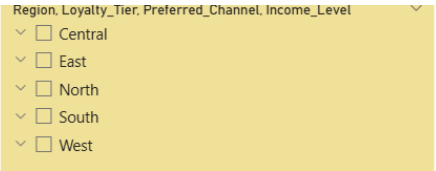
Preferred_Channel	Online		Store		**Total**	
Region	Sum of Amount	Repeat Purchase Rate %	Sum of Amount	Repeat Purchase Rate %	**Sum of Amount**	**Repeat Purchase Rate %**
Central	38,922.18	23.0%	32,960.47	20.4%	**71,882.65**	**21.8%**
East	45,910.19	31.0%	49,025.55	25.0%	**94,935.74**	**28.0%**
North	37,120.21	18.3%	38,394.75	15.8%	**75,514.96**	**17.1%**
South	78,357.03	42.7%	79,281.98	34.0%	**1,57,639.01**	**38.5%**
West	59,535.56	31.0%	56,752.60	29.8%	**1,16,288.16**	**30.4%**
Total	**2,59,845.17**	**83.6%**	**2,56,415.35**	**87.3%**	**5,16,260.52**	**85.4%**

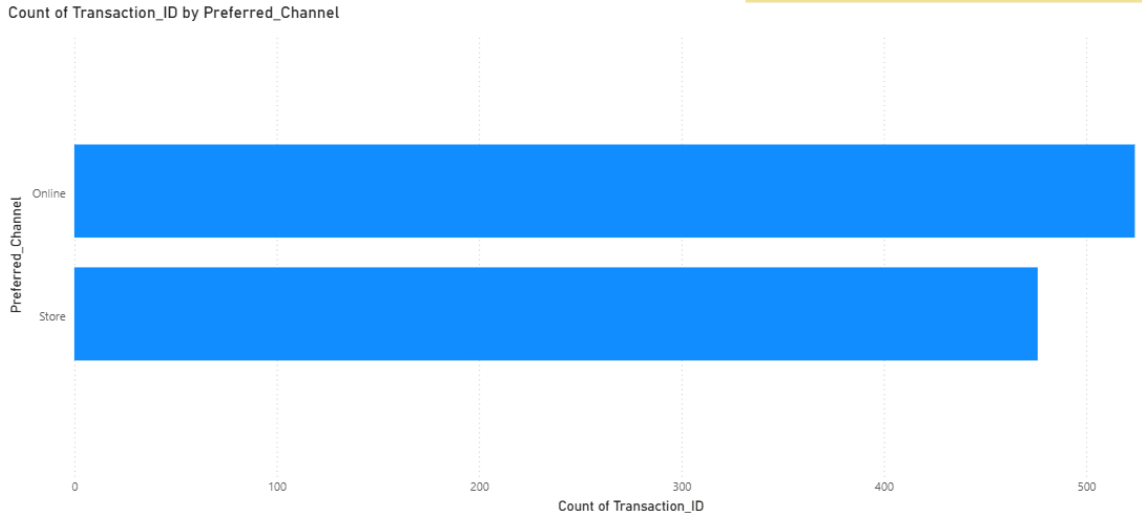
Cross-Channel Analysis: Uses a Pivot Table (Matrix) to cross-reference Region and Preferred Channel against revenue and repeat purchase rates.

Total Revenue by Region

Geographic Revenue : A Map visual that plots Total Revenue by Region, allowing executives to see spatial sales distribution instantly.

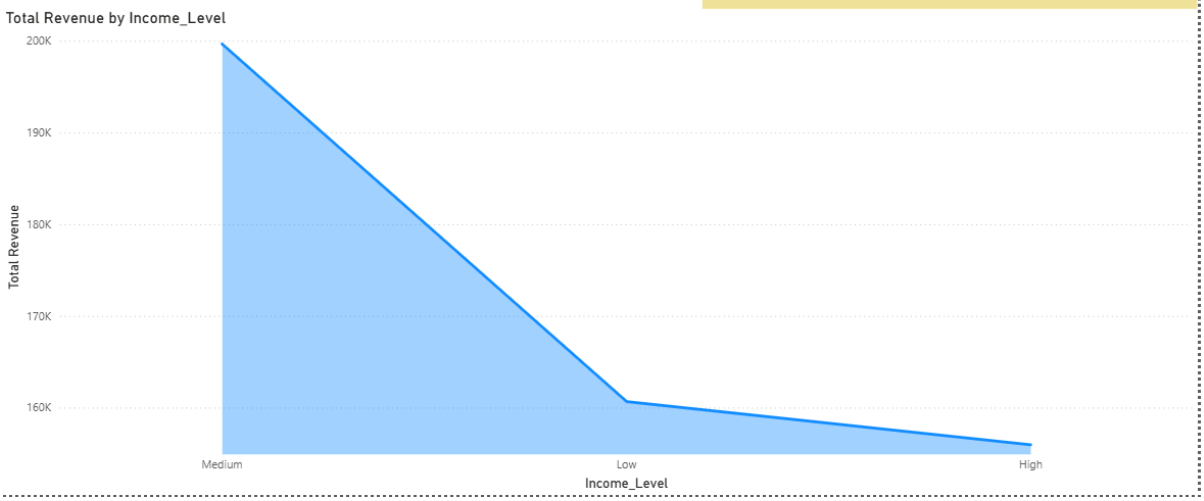
Volume by Channel: A Clustered Bar Chart counting Transaction IDs by Preferred Channel to measure operational load and channel popularity.





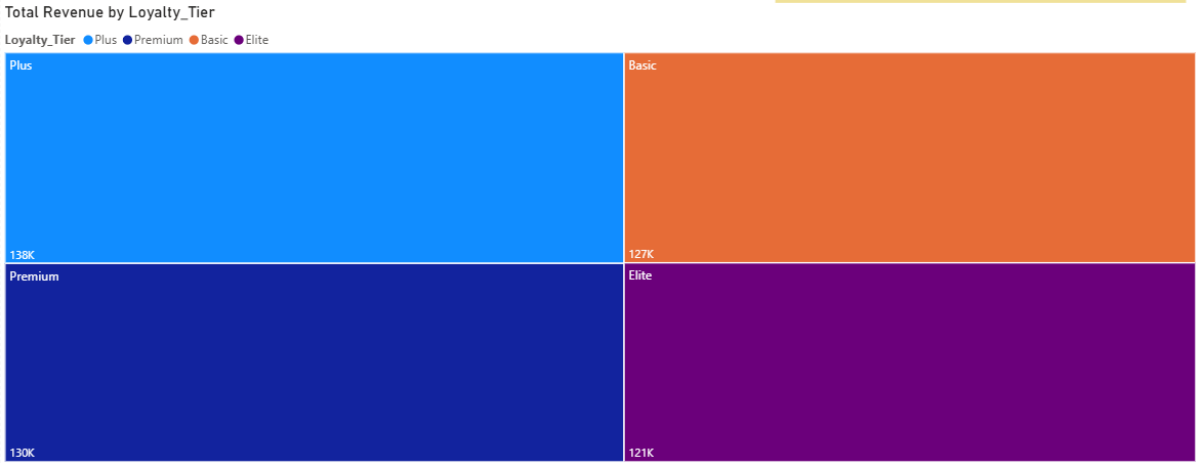
Income Distribution: An Area Chart showing the distribution of Total Revenue across different Income Levels, identifying which economic segments are the biggest spenders.

Region, Loyalty_Tier, Preferred_Channel, Income_Level
- Central
- East
- North
- South
- West



Loyalty Contribution: A Treemap that provides a proportional view of Total Revenue by Loyalty Tier, emphasizing the dominance of high-tier members.

Region, Loyalty_Tier, Preferred_Channel, Income_Level
- Central
- East
- North
- South
- West



Loyalty_Tier	Online	Store	Total
Basic	66905	60357	**127262**
Elite	69079	51473	**120552**
Plus	64531	73691	**138222**
Premium	59330	70894	**130224**
Total	**259845**	**256415**	**516261**

Region, Loyalty_Tier, Preferred_Channel, Income_Level
- Central
- East
- North
- South
- West

Segment Matrix : A detailed Pivot Table that breaks down Total Revenue by Loyalty Tier and Preferred Channel for granular tactical planning.

TOP 3 RECOMMENDATIONS FOR WALMART

---Prioritize "High-CLV, High-Recency" Customers
CLV Analysis and Executive Dashboard provide the perfect roadmap for identifying at-risk value.
Target: Identify customers in the "High CLV Tier" who also have a high value for "Days Since Last Purchase"
Why: These are your historically most profitable customers who are currently showing signs of churn. Because their Lifetime Value is already high, the cost to "win them back" is significantly lower than the cost of acquiring a new high-value customer.

---Loyalty: Bridge the "Earned vs. Redeemed" Engagement Gap
Promotional Impact and Loyalty Contribution indicates a disconnect in how points are used.
Target: Focus on the "Plus" and "Basic" tiers that have a high "Total Points Earned" but low "Total Points Redeemed"
Why: Lower tiers often means the rewards aren't attainable or relevant. If these customers don't feel the "burn" of their points, they have no switching cost and are more likely to churn.

--- Identify "Region-Specific" Underperformers
Target: Look for Regions where the "Repeat Purchase Rate %" is high for "In-Store" but significantly lower for "Online/Grocery Pickup."
Why: If a region has high in-store loyalty but low online adoption, it indicates a friction point in the digital experience (e.g., poor UI, limited pickup slots, or delivery delays) rather than a lack of brand loyalty.

```
Repeat Purchase Rate % =
VAR TotalCustomers = DISTINCTCOUNT('Walmart Customer_Transactions'[Customer_ID])
VAR RepeatCustomers = COUNTROWS(FILTER(VALUES('Walmart Customer_Transactions'[Customer_ID]),
CALCULATE(COUNT('Walmart Customer_Transactions'[Customer_ID])) > 1))
RETURN
DIVIDE(RepeatCustomers, TotalCustomers, 0)
```

```
Average Revenue Per User = [Total Revenue]/[Total Customers]
```

```
Loyalty % = DIVIDE([Total Revenue],CALCULATE([Total Revenue],ALL('Walmart Loyalty_Program'
[Loyalty_Tier])))
```